

B.Tech. DEGREE EXAMINATION, NOVEMBER 2010.

HEAT AND MASS TRANSFER

PART A - (10 x 2 = 20 marks)

- 1) State and explain Fourier law of heat conduction.
- 2) Differentiate between steady state and transient heat conduction.
- 3) Explain the significance of a Grashof number.
- 4) Sketch the hydro dynamic and thermal boundary layer when heating starts at the leading edge.
- 5) Mention the purpose of radiation shield.
- 6) State Kirchoff's and Wein's law of radiations.
- 7) Sketch the temperature profile of a counter flow heat exchanger.
- 8) State the assumptions made while deriving the LMTD formula.
- 9) Compare the mass transfer with heat transfer.
- 10) Explain Fick's law of diffusion.

PART B - (5 x 11 = 55 marks)

UNIT I

- 11) The door of a cold storage plant is made from two 6 mm thick glass sheets separated by a uniform air gap of 2 mm. The temperature of air inside the room is - 20°C and the ambient air temperature is 30°C. Assume heat transfer coefficient between air and glass is 23.26 W/m²K, $k_{\text{air}} = 0.02$ and $k_{\text{glass}} = 0.75$ W/mK. Determine the heat leakage into the room per unit area of the door. Also determine the temperature at the inner surface of the glass.

[or]

- 12) A motor body has an outer diameter of 50 cm, length 40 cm and surface temperature 60°C has 40 longitudinal fins with thickness 8 mm, height 20 mm. Determine the increase in heat flow rate with $h = 23$ W/m²K, $k = 55$ W/mK. Assume ambient temperature 25°C.

UNIT II

- 13) (a) Explain the significance of Prandtl and Nusselt number. (4)
(b) Air at 27°C and 1 atmospheric pressure flows over a flat plate at a speed of 2 m/s. Calculate the heat transfer coefficient in at the first 0.4 m of length of plate by assuming unit depth in z-direction. The plate is heated over its entire length to a temperature of 60°C. (7)

[or]

- 14) Discuss the various regions of flow boiling inside a tube.

UNIT III

- 15) Two large parallel plates with emissivities 0.6 and 0.8 are at 1000 K and 500 K respectively Determine the percentage reduction in heat transfer by placing a shield of reflectivity 0.9 between the plates Also find the shield temperature.

[or]

- 16) (a) Explain the significance of safe factor (5)

- (b) State and prove the Reciprocal theorem of radiation. (6)

UNIT IV

- 17) Derive the LMTD formula for parallel flow heat exchanger.

[or]

- 18) A one Shell two tube pass steam condenser consists of 3000 brass tubes of 20mm diameter. Cooling water enters the tube at 20°C with mass flow rate of 3000 kg/s. The heat transfer coefficient for condensation on the outer surfaces is 15,500 W/m²K. If the heat load of the condenser is 2.3×10^8 W, when the steam condenses at 50°C. Determine

- a) Outlet temperature of cooling water
- b) Overall heat transfer coefficient
- c) Tube length per pass using NTU method.

UNIT V

- 19) (a) Explain the concept of equimolar counter diffusion. (3)

- (b) Hydrogen and oxygen are in equimolar counter diffusion. The total pressure is 1 atm and the temperature is 20°C, $D_{\text{ab}} = 0.41$ m²/s. Partial Pressures at the two planes perpendicular to the direction of diffusion and 95 cm apart are 120 mm and 40 mm of mercury. Calculate the rate of diffusion of mixture in kg-mole/ m²/hr. (8)

[or]

- 20) Estimate the diffusion rate of water at 27°C from the bottom of a test tube of 20 mm in diameter and 4 cm long into dry air at 27°C. Take the diffusion coefficient of water in air as 0.26×10^{-4} m²/s.

B.Tech. DEGREE EXAMINATION, APRIL/MAY 2014

HEAT AND MASS TRANSFER

PART A - (10 x 2 = 20 marks)

- 1) State the Fourier law of conduction.
- 2) Define the term "Thermal Conductivity".
- 3) What is meant by "Free convection"? Explain
- 4) What is meant by "Dropwise condensation"?
- 5) What is a "black body"? Explain.
- 6) Define "Emissivity" of a grey body.
- 7) Differentiate clearly between a counter flow heat exchanger and a parallel flow heat exchanger.
- 8) What is meant by Log Mean Temperature difference (LMTD)?
- 9) State the "Fick's law of Diffusion".
- 10) What is Mass transfer by convection? Explain.

PART B - (5 x 11 = 55 marks)

UNIT I

11) A furnace wall is made up of three layers, one of fire brick, one of insulating brick and one of red brick. The inner and outer surfaces are at 870° C and 40° C respectively. The respective co-efficient of normal conductivities of the layers are 1.0, 0.12 and 0.75 W/m. K and the thickness are 22 cm, 7.5 cm and 11 cm. Assuming close bonding of the layer at their interfaces, find the rate of heat loss per sq. meter per hour and their interface temperature.

[or]

12) Explain briefly the Electrical analogy for solving the conduction heat transfer problems.

UNIT II

13) One hundred tubes of 12 mm in diameter are arranged in a square array and are exposed to steam at atmospheric pressure. Calculate the mass of steam condensed per unit length of tube if the tube wall temperature is maintained at 98° C. Take the properties of water film from the following:

$\rho = 900 \text{ kg/m}^3$, $\mu = 282 \times 10^{-6} \text{ kg/ms}$, $k = 0.61 \text{ W/mK}$, $h_{fg} = 2255 \text{ kJ/kg}$, $T_s = 100^\circ\text{C}$.

[or]

14) Air at 27° C at atmospheric pressure flows over a flat plate at a speed of 2 m / sec. If the plate is maintained at 93° C. Calculate the heat transfer per unit width

of the plate assuming the length of the plate along the flow of air is 2 meters. The properties of air at mean temperature of 60° C are: $\rho = 1.06 \text{ kg / m}^3$, $C_p = 1008 \text{ J /kg-k}$, $k = 0.0285 \text{ W/ m.K}$, $\mu = 20.03 \times 10^{-6} \text{ kg/m- sec}$, $\nu = 18.90 \times 10^{-6} \text{ m}^2/\text{sec}$.

UNIT III

15) A hot water radiator of overall dimensions 2 x 1 x 0.2 m is used to heat a room at 18° C. The surface temperature of the radiator is 60°C and its surface is black. The actual surface area of the radiator is 2.5 times the area of the envelope for convection and the convection heat transfer coefficient is given by $h_c = 1.3(\theta)^{1/3} \text{ W/m}^2\text{K}$. Calculate the rate of heat loss from the radiator by convection and radiation.

[or]

16) (a) Explain Grey Body and emissive power of Grey body in detail. (6)

(b) Explain the following laws of radiation. (Any 2)

Planck law of radiation.

Stefan Boltzmann law.

Kirchoff's law of radiation. (5)

UNIT IV

17) Find the length of the tube required for the following heat transfer where air is heated by exhaust gases : Q (heat transfer) = 8000 watts. Inside diameter (di) and outside diameter (do) of tubes are 5 cm and 6 cm respectively, (hi) Inside heat transfer coefficient = 100 W/m² k. ho (outside heat transfer coefficient, gas side) = 160 W/m² k. Thi = 35° C; Tho = 150° C; Tci = 50° C; Tco = 100° C. Neglect the tube resistance and assume flow arrangement is parallel.

[or]

18) Steam is being condensed on a series of tubes through which cold water is flowing. Each tube is 25 mm in diameter and overall heat transfer coefficient based on outside area is 200 W / m² - k. Water in each tube flows at a rate of 200 kg / hr and entering temperature of 15° C and leaving temperature of 80° C. If the steam is condensing at 100° C. Find the steam condensed on each tube per hour and the length of the tube. Use NTU method.

B.Tech. DEGREE EXAMINATION, NOVEMBER 2014
HEAT AND MASS TRANSFER
PART A - (10 x 2 = 20 marks)

- 1) What is conduction?
- 2) Define thermal conductivity.
- 3) Define convection.
- 4) Define Nusselt Number (Nu).
- 5) State kirchoffs law of radiation.
- 6) What is meant by shape factor?
- 7) What is meant by parallel flow heat exchanger?
- 8) What is meant by effectiveness?
- 9) State Fick's law of diffusion.
- 10) Give two examples of convective mass transfer.

PART B - (5 x 11 = 55 marks)

UNIT I

- 11) Derive the general heat conduction equation for a hollow cylinder.

[or]

- 12) A 10 cm diameter apple approximately spherical in shape is taken from a 20°C environment and placed in a refrigerator where temperature is 5°C and average heat transfer coefficient is 6 W/m²K. Calculate the temperature at the centre of the apple after a period of 1 hour. The physical properties of apple are density = 998Kg/m³, specific heat = 4180 J/Kgk, Thermal conductivity = 0.6 W/mK.

UNIT II

- 13) Air at 30°C, 0.2m/s flow across a 120W electric bulb at 130°C. Find heat transfer and power lost due to convection if bulb diameter is 70mm.

[or]

- 14) Briefly explain the following:
- a) Pool boiling.
 - b) Nucleate boiling.
 - c) Film boiling.

UNIT III

- 15) A black body of 1200 cm² emits radiation at 1000K. Calculate the following
(a) Total rate of energy emission (b) Intensity of normal radiation, (c)

Wavelength of maximum monochromatic emissive power, (d) Intensity of radiation along a direction 60°C to the normal.

[or]

- 16) Two large parallel plates are maintained at a temperature of 600 K and 900 K and emissivities of 0.4 and 0.7 respectively. Determine heat transfer by radiation and also calculate percentage of reduction in heat transfer and shield temperature when another plate of emissivity 0.05 introduced in between them.

UNIT IV

- 17) In a parallel flow heat exchanger, hot water is cooled from 80°C to 40°C by cold water entering at 20°C. The mass flow rate of hot water is 0.2 Kg/s and the mass flow rate of cold water is 0.5 Kg/s. If the individual heat transfer co-efficient on both sides are 600W/m²K. Find the area of heat exchanger.

[or]

- 18) In a counter flow heat exchanger, water at 20°C flowing at the rate of 1200 Kg/hr. It is heated by oil of specific heat 2100 J/kgK flowing at the rate of 520 Kg/hr at the inlet temperature of 95°C Determine (a) Total heat transfer (b) Outlet Temperature of water (c) Outlet temperature of oil.

UNIT V

- 19) Helium diffuses through a plane membrane of 2 mm thick. At the inside the concentration of helium is 0.025 Kg.mole/m³. At the outer side concentration of helium is 0.007 Kg.mole/m³. What is diffusion flux of helium through the membrane? Assume diffusion coefficient of helium with respect to plastic is 1x10⁻⁹ m²/s.

[or]

- 20) Dry air at 30°C and one atmospheric pressure flows over a flat plate of 600 mm long at a velocity of 55 m/s. Calculate the mass transfer coefficient at the end of the plate.

B.Tech. DEGREE EXAMINATION, APRIL/MAY 2015

HEAT AND MASS TRANSFER

PART A - (10 x 2 = 20 marks)

- 1) Define Fourier's law of conduction.
- 2) What is lumped system analysis? When is it used?
- 3) Define the term thermal boundary layer.
- 4) Why heat transfer coefficient for natural convection is much lesser than that for forced convection.
- 5) Distinguish the pool boiling from forced convection boiling.
- 6) Define emissivity.
- 7) Define LMTD of a heat exchanger.
- 8) What is a compact heat exchanger?
- 9) What is Schmidt number?
- 10) State Fick's law of diffusion.

PART - (5 x 11 = 55 marks)

UNIT I

- 11) (a) Derive the general heat conduction equation in Cartesian co-ordinates. (7)
- (b) Explain the mechanism of heat conduction in solids and gases. (4)

[or]

- 12) Aluminum fins, 1.5 cm long and 1 mm thick are placed on a 2.5 cm diameter tube to dissipate heat. The tube surface temperature is 100°C and the ambient temperature is 25°C. Find the heat loss per fin if the heat transfer coefficient between the fin surface and the ambient is 65 W/m²K. Thermal conductivity for the aluminum 200 W/m K. (11)

UNIT II

- 13) (a) Find the convective heat loss from a radiator 0.6 m wide and 1.2 m high maintained at a temperature of 90° C in a room at 14° C. Consider the radiator as a vertical plate. (7)
- (b) Define velocity boundary layer and thermal boundary layer. (4)

[or]

- 14) (a) Explain the various regions of flow boiling in detail. (7)
- (b) Explain nucleate boiling. (4)

UNIT III

- 15) A truncated cone has top and bottom diameters of 10 cm and 20 cm and a height of 10 cm, Calculate the shape factor between the top surface and the side and also the shape factor between the side and itself. (11)

[or]

- 16) A pipe carrying steam having an outside diameter of 20 cm runs in a large room and is expose to air at a temperature of 30°C. The pipe surface temperature is 400°C. Calculate the loss of heat to surroundings per meter length of pipe due to thermal radiation. The emissivity of pipe surface is 0.8. (11)

UNIT IV

- 17) (a) Explain fouling in heat exchangers. (6)
- (b) Classify and compare various types of heat exchangers. (5)

[or]

- 18) The flow rates of hot and. cold water streams running through a parallel flow heat exchanger are 0.2 kg/s and 0.5 kg/s respectively. The inlet temperatures on the hot and cold sides are 75°C and 25 °c respectively. The exit temperature of hot water is 45°C. If the individual heat transfer coefficients on both sides 650 W/m°C. Calculate the area of the heat exchanger. (11)

UNIT V

- 19) (a) Discuss the similarity between Phenomena of heat and mass transfer. (5)
- (b) Discuss diffusive and connective mass transfer. (6)

[or]

- 20) Estimate the diffusion rate of water from the bottom of a test tube 10 mm in diameter and 15 cm long into dry atmospheric air at 25° C. Take the diffusion coefficient of water through air as 0.255 x 10⁻⁴ m²/s. (11)

**B.Tech. DEGREE EXAMINATION, NOVEMBER 2015
(FROM 2009-2012 BATCH)**

HEAT AND MASS TRANSFER

PART A - (10 x 2 = 20 marks)

PART-A

1. Define steady state heat conduction.
2. What is effective thermal conductivity.
3. Differentiate between free and forced convection process.
4. What is meant by nucleate boiling.
5. State “Stefan Boltzman Law”.
6. What is radiation shields? State their names.
7. Sketch a four tube pass and two shell pass heat exchanger.
8. What is meant by effectiveness of a heat exchanger.
9. Define Fourier number
10. What is the significance of Lewis number.

PART - (5 x 11 = 55 marks)

UNIT I

- 11) What is meant by one dimensional heat conduction? Also state the difference between steady state and transient state heat conduction.

[or]

- 12) The thermal conductivity of a material varies linearly with temperature: derive the one-dimensional, steady state heat-conduction equation with internal heat generation by writing the energy balance for a differential volume element in spherical coordinate system

UNIT II

- 13) Brief in detail about velocity boundary layer development on a flat plate and thermal boundary layer on an isothermal flat plate.

[or]

- 14) The bottom of a copper pan 0.32m in diameter is maintained at 120°C by an electric heater. Estimate the power required to boil water in this pan. What is the evaporation rate? Also estimate the critical heat flux.

UNIT III

- 15) Brief about intensity of radiation.

[or]

- 16) A pipe carrying steam having an outside diameter of 20cm runs in a large room and is exposed to air at a temperature of 30°C. The pipe surface temperature is 420°C, calculate the loss of heat to surroundings per meter length of pipe due to thermal radiation. The emissivity of the pipe surface is 0.8. What would be the loss of heat due to radiation if the pipe is enclosed in a 40cm diameter brick conduit of emissivity 0.91.

UNIT IV

- 17) Brief about parallel flow heat exchanger and counter flow heat exchanger. Using simple sketch explain about their temperature distribution.

[or]

- 18) Water enters a cross flow heat exchanger(both the fluids are unmixed) at 10°C and flows at the rate of 4600 kg/h to cool 4000 kg/h of air that is initially at 42°C. Assume the U value to be 150 W/m² K. For an exchanger surface area of 28 m², calculate the exit temperature of water and air. The specific heat of water is 4180J/kg K and air is 1010 J/kg K.

UNIT V

19) Derive the species conservation equation in terms of molar units.

[or]

20) The dry bulb and wet bulb temperature recorded by a thermometer in moist air 26°C and 17°C respectively. Determine the specific humidity of air by assuming the following values.

$Pr = 0.74$, $Sc = 0.6$, $C_p = 1.004 \text{ KJ/Kg K}$.

**B.Tech. DEGREE EXAMINATION, NOVEMBER 2015
(FROM 2013-2014 BATCH ONWARDS)**

HEAT AND MASS TRANSFER

PART A - (10 x 2 = 20 marks)

1. State Fourier's law of conduction.
2. Define critical thickness of insulation with its significance.
3. Define Nusselt number (Nu).
4. Define boundary layer thickness.
5. State Planck's distribution.
6. What is black body and gray body.
7. What is heat exchanger.
8. What is meant by LMTD.
9. List out the various modes of mass transfer.
10. State Fick's law of diffusion.

PART - (5 x 11 = 55 marks)

UNIT I

11) A steel tube ($k = 43.26 \text{ W/m K}$) of 5.08 cm inner diameter and 7.62 cm outer diameter is covered with 2.5 cm layer of insulation ($k=0.208 \text{ W/m K}$) the inside surface of the tube receives heat from hot gas at the temperature of 316°C with heat transfer coefficient of $28 \text{ W/m}^2 \text{ K}$, while the outer surface exposed to the ambient air at 30°C with heat transfer co-efficient of $17 \text{ W/m}^2 \text{ K}$. Calculate heat loss for 3 m length of the tube.

[or]

12) An Aluminium alloy fin of 7 mm thick and 50 mm long protrudes from a wall, which is maintained at 120°C . the ambient air temperature is 22°C . the heat transfer coefficient and conductivity of the fin material are $140 \text{ W/m}^2 \text{ K}$ and $55 \text{ W/m}^2 \text{ K}$ respectively. Determine

- (a) Temperature at the end of the fin
- (b) Temperature at the middle of the fin
- (c) Total heat dissipated by the fin.

UNIT II

13) In a surface condenser, water flows through staggered tubes while the air is passed in cross flow over tubes. The temperature and velocity of air are 30°C and 8 m/s respectively. The longitudinal and transverse pitches are 22mm and 20mm respectively. The tube outside diameter is 18 mm and tube surface temperature is 90°C . Calculate the heat transfer coefficient.

[or]

14) (a) A metal plate 0.609 m high forms the vertical wall of an oven and is at a temperature of 161°C . Within the oven, air at a temperature of 93°C and one atmosphere. Assuming that natural convention conditions hold near the plate. Estimate the mean heat transfer coefficient and rate of heat transfer per unit width of the plate.

(b) A 10 mm diameter spherical steel ball at 260°C is immersed in air at 90°C . Estimate the rate of convective heat loss.

UNIT III

15) A Calculate the following for an industrial furnace in the form of a black body and emitting radiation at 2500°C . (a) Monochromatic emission power at $1.2 \mu\text{m}$ length. (b) Wave length at which the emission is maximum. (c) Maximum emissive power. (d) Total emissive power, and (e) Total emissive power of the furnace if it is assumed as a real surface with emissivity equal to 0.9.

[or]

16) A 12mm outside diameter pipe carries a cryogenic fluid at 90 K. Another pipe of 15mm outside diameter and at 290 K surrounds it coaxially and the space between the pipes is completely evacuated (a) determine the radiant heat flow for 3.5m length of pipe if the surface emissivity for both surface is 0.25 (b) Calculate the percentage reduction in heat flow if a shield of 13.5mm diameter and 0.06 surface emissivity is placed between pipes.

UNIT IV

17) Derive an expression for LMTD in parallel flow and counter flow heat exchangers.

[or]

18) The outside surface of a vertical tube which is 1m long and has an outer diameter of 80mm is exposed to saturated steam at atmospheric pressure and is maintained at 50°C by the flow of cool water through the tube. What is the rate of heat transfer to coolant and what is the rate at which steam is condensed at the surface?

UNIT V

19) A vessel contains a binary mixture of oxygen and nitrogen with partial pressure in the ratio 0.21 and 0.79 at 15°C. The total pressure of the following mixture is 1.1bar. Calculate the following. (a) Molar concentration (b) Mass densities (c) Mass fractions (d) Molar fractions of each species.

[or]

20) Air is contained in a tyre tube of surface area 0.5 m² and wall thickness 10mm. The pressure of air drops from 2.2 bar to 2.18 bar in a period of 6 days. The solubility of air in the rubber is 0.72m³ of air per m³ of rubber at 1 bar. Determine the diffusivity of air in rubber at the operating temperature of 300 K, if the volume of air in the tube is 0.028m³.

B.Tech. DEGREE EXAMINATION, NOVEMBER 2016

(FROM 2009-2012 BATCH)

HEAT AND MASS TRANSFER

PART A - (10 x 2 = 20 marks)

1. Define the term thermal diffusivity.
2. What are the uses of Heisler's charts?
3. What are the significance of Prandtl number?
4. How to define the Reynold's number in condensation? What is its critical value?
5. State the kirchoff's law of radiation.
6. What are the factors affecting the emission of radiation?
7. Draw the temperature variation profile for parallel flow and counter flow heat exchangers.
8. What do you understand about the effectiveness of heat exchanger?
9. What is the physical significance of Schmidt number?
10. What are the limitations of Fick's law?

PART - (5 x 11 = 55 marks)

UNIT I

11) A boiler furnace has the effectiveness dimensions 4m by 3m by 3m high. The walls are constructed from an inner firebrick wall 25 cm thick ($k = 0.4 \text{ W/mK}$), a layer of ceramic blanket insulation ($k = 0.2 \text{ W/mK}$) 8cm thick and a steel protective layer ($k = 54 \text{ W/mK}$) 2mm thick. The inside temperature of the firebrick layer was measured as 600°C and the temperature of the outside of insulation as 60°C . Determine the rate of heat loss through the vertical walls of the furnace. Also calculate the temperature drop across the steel layer.

[or]

12) Determine the thermal conductivity of long metal rod of diameter

of 3.2cm, its one end is kept at a hot source with large portion of it projecting into the room, where temperature is maintained at 25°C after steady state attains, the temperature between the two points, 15cm apart are found to be 175°C and 120°C respectively. The convective heat transfer coefficient between the metal rod and the atmospheric air is $30 \text{ W/m}^2\text{K}$. Determine the thermal conductivity of the rod material.

UNIT II

13) Water enters into a tube of 60mm diameter at 20°C and leaves at 80°C . If the tube receives heat at a rate of 2000 W/m^2 at the surface, determine the surface temperature at the outlet of the tube and the length of the tube for a flow rate of 0.01 kg/s .

[or]

14) (a) Differentiate between the Filmwise and Dropwise condensation.

(b) Explain the various regimes of boiling.

UNIT III

15) (a) What is Stefan-Boltzmann's law? Explain the concept of total emissive power of a surface.

(b) Describe the concept of black body.

[or]

16) Two parallel discs of diameter 1cm (thickness negligible) which are separated by a distance of 0.7cm, located in a large room. The room walls are maintained at a temperature of 310K . One disc is maintained at a temperature of 1200K and the other one at 600K , the emissivities of the plates are 0.25 and 0.5 respectively. If the plates

exchange heat between themselves and surroundings, find the net heat transfer to each disc and the room. Assume only the disc surfaces facing each other.

UNIT IV

17) A heat exchanger is required to cool 55000 kg/hr of an oil from 120°C to 60°C using 40000 kg/hr of water at 25°C. Calculate the surface area of heat exchanger required for parallel flow and counter flow mode. Take $U = 600 \text{ W/m}^2\text{K}$, $C_{p_w} = 4186 \text{ kJ/kgK}$ and $C_{p_{oil}} = 5200 \text{ kJ/kgK}$.

[or]

18) In a chemical processing plant water at the rate of 13260 kg/h is heated from 38°C to 55°C in tubes with 6810 kg/h hot water (on the shell side) entering the exchanger at 94°C. The average velocity of water in the tubes is 36.6 cm/s, tube diameter is 1.9 cm and the overall heat transfer coefficient is $1420 \text{ W/m}^2\text{K}$. If the floor space available cannot accommodate tube longer than 2.44m, calculate the number of tube passes, number of tubes per pass and the length of the tubes.

UNIT V

19) (a) Briefly describe the various modes of mass transfer.

(b) List out the applications of mass transfer.

[or]

20) Calculate the rate of diffusion of water vapour from a pool of water at the bottom of a well which is 6m deep and 2.5m diameter to dry ambient air over the top of the well. The entire system may be assumed at 30°C and one atmospheric pressure. The diffusion coefficient is $2.5 \times 10^{-5} \text{ m}^2/\text{s}$.

**B.Tech. DEGREE EXAMINATION, NOVEMBER 2016
(FROM 2013-2014 BATCH ONWARDS)**

HEAT AND MASS TRANSFER

PART A - (10 x 2 = 20 marks)

1. State Fourier's law of heat conduction.
2. Write the difference between fin efficiency and fin effectiveness.
3. How does laminar flow differ from turbulent flow.
4. Write about critical heat flux.
5. State Kirchoff's law of radiation.
6. What is grey body.
7. What is log mean temperature difference.
8. What are the parameters affecting the fouling.
9. What do you mean by mass transfer.
10. Define the term molecular diffusion.

PART - (5 x 11 = 55 marks)

UNIT I

11) Calculate the rate of heat loss for a red brick wall of length 5m, height 4m and thickness 0.25m. The temperature of the inner surface is 120°C and that of the outer surface is 40°C. The thermal conductivity of red brick, $k = 0.70 \text{ W/m K}$. Also calculate the temperature at an interior point of the wall 0.20 m distant from the inner wall.

[or]

- 12) (a) Write a note on different fin profiles.
- (b) An Aluminium alloy fin of 5mm thick and 40mm long protrudes from a wall. The base temperature is 420°C and ambient air temperature is 25°C. The heat transfer coefficient between aluminum rod and environment is $25 \text{ W/m}^2 \text{ K}$. Calculate the heat loss from the fin of material taking its thermal conductivity as 200 W/m K .

UNIT II

13) (a) Discuss briefly the development of velocity boundary layer for flow through a pipe.

(b) Air at 30°C flows over a flat plate at a velocity of 2m/s. The plate is 2m long and 1.5m wide. Calculate the

(i) Hydro dynamic and thermal boundary layer thickness at the trailing edge of the plate

(ii) Total drag force

[or]

14) Water is to be boiled at atmospheric pressure in a mechanically polished stainless steel pan placed on the top of a heating unit. The inner surface of the bottom of the pan is maintained at 108°C. The diameter of the bottom of the pan is 0.30m. Assuming $C_{SF} = 0.0130$, Calculate the following:

- (a) The rate of heat transfer to the water
- (b) The rate of evaporation of water.

UNIT III

15) Find the relative heat transfer between two large planes at temperature 1000K and 500K when they are

- (a) Black bodies
- (b) Grey bodies with emissivities of each surface is 0.7

[or]

16) Emissivities of two large parallel plates maintained at 800°C and 300°C are 0.3 and 0.5 respectively. Find the net radiant heat exchanger per square metre of the plates, if a polished aluminium shield ($\epsilon = 0.05$) is placed between them. Find also the percentage of reduction in heat transfer.

UNIT IV

17) In a counter flow, single pass heat exchanger is used to cool the engine oil from 150°C to 55°C with water, available at 23°C as the cooling medium. The specific heat of oil is 2125J/Kg K . The flow rate of cooling water through the inner tube of 0.4m diameter is 2.2 Kg/s . The flow rate of oil through the outer tube of 0.75m diameter is 2.4 Kg/s . If the value of the overall heat transfer coefficient is $240\text{W/m}^2\text{ K}$, how long must the heat exchanger be to meet its cooling requirement?

[or]

18) Water enters a cross flow heat exchanger (both fluids unmixed) at 5°C and flows at the rate of 4600 kg/hr to cool 4000 kg/hr of air that is initially at 40°C . Assume the U value to be $150\text{ W/m}^2\text{ K}$, for an exchanger surface area of 25 m^2 , calculate the exit temperature of air and water. (Use NTU method)

UNIT V

19) Air at 10°C with a velocity of 3m/s flows over a flat plate. If the plate is 0.3m long, calculate the mass transfer co-efficient.

[or]

20) Air at 1atm , 25°C containing small quantities of Iodine flows with velocity of 5.18m/s inside a 3.048 cm diameter tube. Determine the mass transfer coefficient for Iodine transfer from the gas stream to the wall surface if C_m is the mean concentration of Iodine in Kg mol /m^3 in the air stream. Determine the rate of deposition of Iodine on the tube surface where the Iodine concentration is Zero.

B.Tech. DEGREE EXAMINATION, APRIL/MAY 2017

(FROM 2009-2012 BATCH)

HEAT AND MASS TRANSFER

PART A - (10 x 2 = 20 marks)

1. Give the mechanism of heat conduction
2. Define thermal conductivity of a material
3. What is forced convection
4. Define thermal resistance
5. Define emissive power
6. What is known as radiation shape factor
7. What is heat exchanger
8. Define number of transfer unit
9. When will the diffusion mass transfer occurs
10. Give two example of mass transfer

PART - (5 x 11 = 55 marks)

UNIT I

11) A wall of 0.6 m thickness having thermal conductivity of 1.2W/mk. the wall is to be insulated with a material having an average thermal conductivity of 0.3 W/mk. inner and outer surface temperature is 1000°C and 10°C respectively. If the heat transfer rate is 1400W/m². Calculate the thickness of insulation. .

[or]

12) A long rod of 5cm diameter, its base is connected to a furnace wall at 150°C while the end is projection into a room at 20°C. The temperature of the rod at a distance of 20cm distance from its base is 60°C. the conductivity of the material is 200W/mk. Determine the convective heat transfer co-efficient

UNIT II

13) Air at 25°C flows over a flat plate at a speed of 5m/s and heated to 135°C. The plate is 3m long and 1.5m wide. Calculate the local heat transfer coefficient at x=0.5m and the heat transferred in the first 0.5m of the plate.

[or]

14) A large vertical plate 5m height is maintained at 100°C and exposed to air at 30°C, calculate the convective heat transfer co-efficient.

UNIT III

15) Two larger parallel plates are maintained at a temperature of 900K and 500K respectively. Each plate has an area of 6m². Compare the net heat exchange between the plates for the following cases : (a) both plate are black (b) plate have an emissivity of 0.5

[or]

16) Two black square plate of size 1*1 m are placed parallel to each other at a distance of 0.4m. One plate is maintained at a temperature of 900°C and the other at 400°C. Find the net heat exchanger of energy due to radiation between the two plates

UNIT IV

17) In a counter flow double pipe heat exchanger water heated from 25°C to 65°C by oil with a specific heat of 1.45 KJ/KgK and mass flow rate is 0.9Kg/s the oil is cooled from 230°C to 160°C if the overall heat transfer co-efficient is 420 W/m²°C, calculate the following : (a) the rate of heat transfer , (b)the mass flow rate of water (c) the surface area of the heat exchanger

[or]

18) It is designed to use double pipe counter flow heat exchanger to cool 3Kg/s of oil ($C_p = 2.1 \text{ KJ/KgK}$) from 120°C . cooling water at 20°C enter the heat exchanger at a rate of 10Kg/s. the overall heat transfer coefficient of the heat exchanger is $600\text{W/m}^2 \text{ k}$ and the heat transfer area is 6m^2 . Calculate the exit temperature of oil and water .

UNIT V

19) Helium diffuses through a plate membrane of 2mm thick. At the inner side the concentration of helium is 0.025Kg mole/m^3 . at the outer side the concentration of helium is 0.007Kg mole/m^3 . What is the diffusion flux of helium through the membrane? Assume diffusion coefficient of helium with respect to plastic is $1 \times 10^{-9} \text{ m}^2/\text{s}$

[or]

20) **CO₂** And air experience equimolar counter diffusion in a circular tube whose length and diameter are 1.2m and 60mm respectively. The system is at a total pressure of 1 atm and a temperature of 273K. The end of the tube is connected to large chamber. partial pressure of **CO₂** at one end is 200mm of Hg. Calculate the following (a) Mass transfer rate of CO (b) mass transfer rate of air

**B.Tech. DEGREE EXAMINATION, APRIL/MAY 2017
(FROM 2013-2014 BATCH ONWARDS)**

HEAT AND MASS TRANSFER

PART A - (10 x 2 = 20 marks)

- 1) Define heat flux.
- 2) What is coefficient of Thermal conductivity?
- 3) Sketch boundary layer development in a circular pipe.
- 4) Define critical heat flux.
- 5) What is Intensity of radiation?
- 6) Define shape factor.
- 7) Write the advantage of NTU method over the LMTD method.
- 8) What is effectiveness of a heat exchanger?
- 9) Give some examples of diffusion mass transfer?
- 10) What is convective mass transfer?

PART - (5 x 11 = 55 marks)

UNIT I

11) A long rod is exposed to air at 298°C. It is heated at one end. At steady state conditions, the temperature at two points along the rod separated by 120mm are found to be 130°C and 110°C respectively. The diameter of the rod is 25mm OD and its thermal conductivity is 116 W/m°C. calculate the heat transfer coefficient at the surface of the rod and also the heat transfer rate.

[or]

12) a composite wall consists of 10cm thick layer of building brick, $k=0.7$ W/mK. An insulating material of $k=0.08$ W/mK is to be added to reduce the heat transfer through the wall by 40%. Find its thickness.

UNIT II

13) Air at 25°C flows over 1m x 3m (3m long) horizontal plate maintained at 200°C at 10m/s. calculate the average heat transfer coefficients for both laminar and turbulent regions. take $Re_{critical}=3.5 \times 10^5$.

[or]

14) A steam pipe 10 cm OD runs horizontally in a room at 23°C. take outside temperature of pipe as 165°C. determine the heat loss per unit length of the pipe. pipe surface temperature reduces to 80°C with 1.5cm insulation. what is the radiation in heat loss?

UNIT III

15) Describe the phenomenon of radiation from real surfaces.

[or]

16) Explain briefly the following :

- (a) reflectivity and transmissivity
- (b) reciprocity rule and summation rule.

UNIT IV

17) It is desired to use a double pipe counter flow heat exchanger to cool 3 kg/s of oil ($C_p=2.1$ kJ/kgK) from 120°C. Cooling water at 20°C enters the heat exchanger at a rate of 10 kg/s. The overall heat transfer coefficient of the heat exchanger is 600 W/m²K and the heat transfer area is 6 m². Calculate the exit temperatures of oil and water.

[or]

18)Define effectiveness of a heat exchanger. Derive an expression for the effectiveness of a double pipe parallel flow heat exchanger. State the assumptions made.

UNIT V

19) (a) Define mass concentration, molar concentration, mass fraction and mole fraction

(b)The diffusivity of CCl_4 in air is determined by observing the steady state evaporation of CCl_4 in a tube of 1 cm diameter exposed to air. the CCl_4 liquid level is 10cm below the top level of the tube. the system is held 25°C and 1 bar pressure. the saturation pressure of CCl_4 at 25°C is 14.76 kPa. If it is observed that the rate of evaporation of CCl_4 is 0.1 kg/hour determine the diffusivity of CCl_4 into air.

[or]

20) (a) What are the assumption made in the T-D transient mass diffusion problems?

(b) Define the non dimensional numbers in mass transfer.
